

Ethan Lien

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EDUCATION

University of California, Los Angeles

Master of Science in Mechanical Engineering: Systems and Controls

Los Angeles, CA

June 2026

University of California, Los Angeles

Bachelor of Science in Mechanical Engineering, Minor in Linguistics

Los Angeles, CA

June 2025

Courses: Kinematics/Dynamics/Control of Robotic Systems, Dynamics and Control of Biological Oscillations, Nonlinear Dynamics Systems, Digital Control and Learning of Physical Systems, Optimal Control

PROJECTS

Bruin Underwater Robotics

bruin-underwater-robotics.github.io

Project Lead

May 2024 – Aug 2025

- Led a team of 25 engineers to create an autonomous underwater vehicle (AUV) for the Robosub competition
- Optimized mechanical and pneumatic designs for underwater performance
- Guided electrical engineering team in designing on-board power systems and communication systems
- Managed financial resources and sponsorships; attained over \$5000 in combined funding and materials
- Ranked 25th out of 60 teams internationally, considering both technical performance and design documentation

Chassis Subteam Lead

Oct 2023 – Aug 2024

- Designed the structural frame of a competition AUV for the Robosub competition
- Directed a team of 9 engineers to develop external mounting to achieve 6 DoF AUV motion and optimize internal electronics configuration for cable management
- Verified initial density calculations and underwater capabilities through submersion testing at depth of 4.5 meters
- Ranked 21st out of 41 teams internationally

DiveOps Subteam Lead

Oct 2022 – Jun 2023

- Designed and fabricated a profiling float for the MATE ROV competition
- Implemented a continuous servo-driven buoyancy engine to enable vertical ascension via alterations in density
- Employed time-keeping and wireless communication through clock and Bluetooth modules for Arduino

EXPERIENCE

Solution Coordinator Intern

Jul 2023 – Aug 2023

Nippon Technology Solution Co., Ltd.

Kobe, Japan

- Worked in a team of engineering interns to assemble TORNADO[®] shrink-wrap packaging machines; constructed and installed conveyor belt and convective heating elements
- Inspected and tuned machine parameters to meet high-precision performance specifications
- Developed insight in maintenance accessibility through troubleshooting during assembly process

Course Reader

Mar 2023 – Jun 2025

UCLA Mechanical & Aerospace Engineering

Los Angeles, CA

- Corrected assignments and exams for *MAE 162A: Introduction to Mechanisms and Mechanical Systems* and *MAE 102: Dynamics of Particles and Rigid Bodies*
- Provided feedback to students to develop fundamental skills for analysis of dynamic mechanical systems

TECHNICAL SKILLS

Software: MATLAB, Simulink, SolidWorks, KiCAD

Libraries: Controls System Toolbox, Optimization Toolbox, Robotic Systems Toolbox, Symbolic Math Toolbox